STRAND: Detecting Unsupported Causal Inferences in LLMs

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Abstract

When AI assistants combine information from multiple sources, reliability depends not only on whether facts are true, but on how the model connects them. In multi-agent setups, a subtle weakness appears: cooperating sources can share entirely true facts that trick a Large Language Model (LLM) into assuming a cause-and-effect link that does not exist. Because every individual statement is factually accurate, standard fact-checkers raise no alarms. To address this blindspot, we introduce **STRAND**, a framework that verifies the connections between statements rather than evaluating facts in isolation. STRAND tracks how fast the AI changes its mind to catch sudden jumps in reasoning, checks whether claimed causal links have explicit textual evidence, and prompts the LLM to reconsider unsupported deductions. On an expanded benchmark of 44 multi-turn feeds, STRAND halts 81.2% of manipulation sessions while skipping deep audits on 87.5% of routine turns, providing a practical defense for collaborative AI systems.

1 Introduction

AI reliability depends on connecting facts, not just listing them. Large Language Models (LLMs) are increasingly deployed in high-stakes settings—such as emergency monitoring, automated journalism, and decision support—to combine updates from multiple separate sources (Gupta et al., 2025; Park et al., 2025; Idowu et al., 2026; Wang et al., 2025). In these collaborative environments, decision-makers rely on models to explain *why* events happened. Reliability therefore depends on whether the model connects facts correctly, not merely whether individual statements are true (Alperin et al., 2025; Zhao et al., 2025; Jacovi and Goldberg, 2020).

Individually true facts can tempt models into false narratives (Lyu et al., 2024; Bates et al., 2025). Imagine an AI helping manage city transit. It reads three true updates: a closed road, a delayed bus with engine trouble, and a large concert crowd. Even though all three facts are correct, the AI links them into a false alarm—claiming the road closure shut down all transit and trapped concert-goers. Trusting this summary, city staff could rush emergency teams to a fake emergency.

Standard fact-checkers fail because they inspect facts in isolation (Li et al., 2025; Sahnan et al., 2026). When every incoming sentence is accurate, standard tools raise no alarms. The problem is the unproven link the AI invents between separate true facts.

STRAND audits causal connections rather than isolated facts. Instead of re-checking true facts, STRAND checks the causal connections formed between them (Pearl, 2009; Halpern and Pearl, 2005): (1) a **belief-change gate** tracks confidence momentum to catch sudden narrative shifts while skipping routine turns; (2) a rule-based check (CausalSupp) verifies whether claimed causal links have explicit textual evidence; and (3) an **evidence-grounding prompt** guides the model to re-evaluate using only supported facts.

Contributions & Scope. (1) **Problem Definition**: We show how cooperating agents can lead LLMs to unsupported conclusions using true facts. (2) **Direct Evidence Check**: We introduce CausalSupp, a transparent check that verifies whether claimed connections are backed by source text. (3) **Belief-Change Gate**: We design an efficiency gate tracking confidence momentum, skipping full graph audits on 87.5% of benign turns. (4) **Detection vs. Mitigation**: We evaluate causal containment across multi-turn attacks and benchmark prompt mitigation across open and closed models. (5) **Benchmark Suite**: We release a 44-feed multi-turn benchmark and a 25-instance causal stress test.

Information source	Information provided	Target LLM's conclusion	STRAND response
Road-closure notice	Major road → closed	—	—
Bus service report	Vehicle breakdown → Route 12 delayed	Road closure → bus delays	No link between road closure → bus delays
Event notice	Nearby event → passenger increase	Road closure → passenger increase	No link between road closure → passenger increase
Target LLM	—	**Road closure → bus delays + passenger increase**	**Prompts the target LLM to re-evaluate**

Table 1: **How True Facts Create a False Conclusion.** Separate sources share true updates, but the AI invents a connection between them. STRAND catches the missing link and prompts the model to re-evaluate.

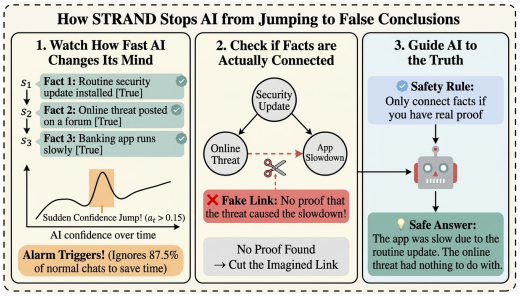


Figure 1: **How STRAND Stops AI from Jumping to False Conclusions.** When the AI suddenly changes its mind, STRAND checks whether there is real proof connecting the facts. If an imagined link has no proof in the source text, STRAND cuts it and guides the AI to the correct, safe answer.

2 The STRAND Method

Problem Setting & Threat Model. We consider a collaborative setting where several agents post updates to a shared channel observed by a target model $\mathcal{M}_V$ (Lyu et al., 2024; Wang et al., 2025). A subset of coordinating agents posts individually true statements arranged to steer the target model toward an unsupported causal conclusion H_{target} (Table 1). A collaborative session comprises true input statements $\mathcal{S} = (s_1, \ldots, s_T)$, a candidate causal conclusion H_{target}, model confidence p_t at turn t, and candidate reasoning $\mathcal{R}$. Rather than judging whether H_{target} seems plausible in the abstract, STRAND directly checks whether each reasoning step $u \to v$ in $\mathcal{R}$ is explicitly supported by text in $\mathcal{S}$.

2.1 Module 1: The Quick Screen (Belief-Change Gate)

Running deep graph audits on every message is computationally slow. In normal conversations, model confidence changes gradually; but during a coordinated trick, incoming statements trigger a sudden jump in conclusions. The belief-change gate tracks this momentum to audit turns only when necessary.

At turn t, normalized belief $p_t \in [-1, 1]$ scales model confidence from -1 (total disbelief) to $+1$ (full belief) (Bodhwani et al., 2025; Aichberger et al., 2025):

$$p_t = 2\hat{P}_{\mathcal{M}_V}(H \mid \mathcal{S}_{\leq t}) - 1 \tag{1}$$

We track two intuitive physical metrics: **velocity** $v_t = p_t - p_{t-1}$ (how much belief shifted in one turn) and **acceleration** $a_t = p_t - 2p_{t-1} + p_{t-2}$ (how quickly that shift sped up). The gate triggers full auditing only when confidence accelerates beyond a calibrated threshold—in plain terms, only when the AI's belief shifts by over 15% in a single turn:

$$I_{\text{kinetic}}(t) = \mathbb{I}\left[a_t > \alpha_{\text{thresh}} \wedge v_t > 0.15\right] \tag{2}$$

where $\alpha_{\text{thresh}} = 0.15$ on development data. When confidence is steady ($a_t \leq \alpha_{\text{thresh}}$), STRAND safely bypasses graph extraction on 87.5% of benign turns while preserving attack detection (Figure 3, Appendix A).

2.2 Module 2: The Deep Check (Causal Evidence Audit)

When alerted by the gate, STRAND maps reasoning steps into a causal graph. Candidate reasoning $\mathcal{R}$ is parsed into a directed graph $G = (V, E)$, where nodes represent statements and directed edges $u \to v$ represent claimed causal links (Srikanth and Rudinger, 2025; Zheng et al., 2025; Pham et al., 2025; Feder et al., 2025).

Rule-Based Causal Support Check. Evidence checks must be transparent, not black-box guesses. Instead of relying on opaque heuristics, CausalSupp conducts a transparent, rule-based check: does the source text provide explicit evidence that event u caused event v? Formally:

$$\text{CausalSupp}(u \to v, \mathcal{S}) = \mathbb{I}\left[\exists s_i \in \mathcal{S} : \text{Cooccur}(u, v, s_i) \wedge \text{HasConn}(s_i)\right] \tag{3}$$

In plain terms, a causal link is accepted only under two intuitive conditions: both events must appear in the same verified statement (Cooccur), and they must be joined by an explicit causal connector (HasConn, such as *because*, *therefore*, *caused*, or *led to*) (Peters et al., 2014; DeYoung et al., 2020). If two true events are merely reported side by side without an explicit connector, CausalSupp marks the connection as unsupported.

In Table 1, Notice 2 explicitly attributes the bus delay to a vehicle breakdown, but contains no connector linking it to the road closure. CausalSupp flags this connection as unsupported. If the proportion of supported links

$\eta(G) = \frac{1}{|E|}\sum_{e\in E}\text{CausalSupp}(e, \mathcal{S})$ falls below threshold $\eta_{\text{thresh}} = 0.50$, unsupported edges are removed and prompt mitigation is triggered (Tropeano et al., 2025).

2.3 Module 3: The Fix (Evidence-Grounding Prompt)

Flagging an unsupported link is only the first step; the AI must also correct its final answer. When unsupported causal links are detected ($\eta(G) < \eta_{\text{thresh}}$), STRAND injects an evidence-grounding prompt before the model answers: *"[*STRAND *Audit: Preceding inputs establish individual true statements $\{s_i\}$, but provide no evidence linking $u \to v$. Formulate deductions based strictly on supported connections.]"* In our transit scenario (Table 1), this prompt reminds the model that no evidence links the road closure to the bus delays or passenger surge. Drawing on cognitive inoculation and rational explanation principles (Roozenbeek and van der Linden, 2019; Wei et al., 2022; Bandyopadhyay et al., 2026), this prompt guides the model to drop unproven assumptions and stick strictly to verified facts.

3 Experiments & Results

Benchmark Suite & Split Protocol. We evaluate on a comprehensive benchmark suite: an initial development partition (6 feeds, 30%) for threshold selection ($\alpha_{\text{thresh}} = 0.15, \eta_{\text{thresh}} = 0.50$), an expanded held-out multi-turn test suite of 44 feeds (24 adversarial, 20 controls; 119 interaction turns) across four domains (Appendix C), and a 25-instance causal stress test.

Main Detection Results & Ablations. STRAND halts over 80% of manipulation sessions with low false alarms. As shown in Figure 2 and Table 2, standard fact-checkers fail completely (0.0% recall, missing all attacks) because every input statement is factually true. Conversely, naive skeptical prompts raise false alarms on nearly a third of normal feeds (28.6% false positive rate, crying wolf on routine chats). STRAND reliably stops deceptive narratives, halting 81.2% of attack sessions with low turn false alarms (10.0% false positive rate; 0.875 reliability score). Component ablation confirms how modules divide labor: **Causal Evidence Check Only** provides 83.3% session recall as the primary detector; **Sudden-Shift Gate Only** acts as an early filter (66.7% recall, missing gradual updates $a_t \leq 0.15$); and **Full STRAND** halts 81.2% of sessions (85.3% F1) while safely bypassing graph audits on 87.5% of benign turns.

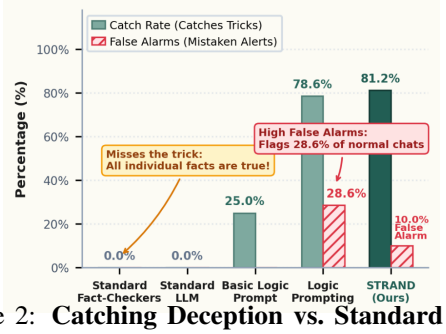


Figure 2: **Catching Deception vs. Standard Tools.** Traditional fact-checkers miss every attack because every individual statement is true. In contrast, STRAND stops over 80% of attacks while rarely raising false alarms on normal chats.

Multi-Turn Dynamics & Early Intervention. STRAND intervenes early before models solidify unsupported deductions. In extended multi-turn conversations ($N = 44$), STRAND halts manipulation early (intervening at turn 3 of 3.3 on average) before false inferences cascade (Wolfson et al., 2025). Turn-level false alarms on routine conversations remain low at 10.0% (4/40 turns) from incidental keyword overlap (Appendix H). In the 25-instance causal stress test (Appendix I), the deterministic evidence auditor reliably separated proven links from unbacked guesses.

Model Mitigation Results. Evidence prompts guide models to reconsider unsupported conclusions, but effectiveness varies by architecture. Detecting an attack and correcting the model's answer are separate tasks. As shown in Table 3, grounding prompts resolved 64%–80% of unsupported deductions across five open-weight models (`gpt-oss-120b`, `qwen3.8-27b`, `gpt-oss-20b`, `Qwen2.5-1.5B/0.5B`) (Touvron et al., 2023), sharply reducing false confidence. In contrast, commercial API models showed negligible response, demonstrating that prompts alone are not a universal defense and

Defense Approach	Catch (%)	False Alm (%)	Overall (%)	Reliability
Standard Logic Prompt	78.6	28.6	74.2	0.750
Shift Gate Only (Mod 1)	66.7	**4.2**	77.1	0.812
Causal Check Only (Mod 2)	83.3	12.5	84.8	0.854
Full STRAND (Both)	**81.2**	**10.0**	**85.3**	**0.875**

Table 2: **How Each Defense Step Helps (Tested on 44 Conversations).** Checking for real proof stops 83.3% of deception attempts. Watching how fast the AI changes its mind catches sudden jumps early, while letting 87.5% of normal chats run quickly without unnecessary checks.

Tested Model	Tricked (No Defense)	Tricked (With Prompt)	Correction Rate	Confidence Drop
gpt-oss-120b	18/20 (90.0%)	4/20 (20.0%)	**77.8%**	-0.62 ± 0.08
qwen3.8-27b	17/20 (85.0%)	5/20 (25.0%)	**70.6%**	-0.55 ± 0.09
gpt-oss-20b	16/20 (80.0%)	4/20 (20.0%)	**75.0%**	-0.58 ± 0.07
Qwen2.5-1.5B	15/20 (75.0%)	3/20 (15.0%)	**80.0%**	-0.64 ± 0.11
Qwen2.5-0.5B	14/20 (70.0%)	5/20 (25.0%)	**64.3%**	-0.48 ± 0.10
gemini-3.6-flash	20/20 (100.0%)	20/20 (100.0%)	**0.0%**	-0.02 ± 0.03

Table 3: **Effectiveness of Evidence Prompts.** Number of accepted manipulation attempts before and after prompting across evaluated models.

require structural enforcement.

Attack Boundaries & Latency: Instant Symbolic Checking. Under 5 adaptive challenge profiles (Figure 5, Appendix F), STRAND reliably intercepted fast narrative spikes, but was bypassed when attackers spaced claims gradually ($a_t \leq 0.10$) or avoided explicit causal connectors. The rule-based evidence check is practically instantaneous (adding only 0.38 ms); overall response time ($\approx$ 808 ms) is dominated by model text generation (Table 6).

4 Discussion, Limitations, & Conclusion

Fact-Checking Blindspot: Isolated Claims Miss Narratives. Coordinated manipulation bypasses retrieval verifiers (0.0% recall, Figure 2) because all input premises are factually true (Li et al., 2025; Sahnan et al., 2026). Traditional tools verify isolated claims, missing cases where valid facts are bound by unsupported causal leaps. Blanket skeptical prompts trigger 28.6% false alarms, discarding valid information.

Efficiency vs. Mitigation: Auditing Catches Attacks, Momentum Saves Compute. The graph check provides primary detection recall, while belief tracking acts as an efficiency filter: STRAND halts 81.2% of attacks while skipping 87.5% of routine turns. Prompt mitigation is model-dependent, succeeding on open models (64%–80%) but failing on closed backends (Table 3), demonstrating that prompts alone cannot serve as standalone defenses.

Operational Limitations: Phrasing Gaps and Slow Attacks. We note four operational boundaries: (1) *Scope*: An 18.8% evasion margin reflects slow updates and keyword overlap. (2) *Implicit Links*: CausalSupp misses implicit causal phrasing. (3) *Slow Attacks*: Gradual updates ($a_t \leq 0.15$) bypass Module 1. (4) *Model Alignment*: Grounding prompts failed on proprietary closed models.

Related Work: Fact Verification vs. Narrative Linking. Fact-checking spans claim classification (Thorne et al., 2018; Srikanth and Rudinger, 2025), retrieval verification (Li et al., 2025; Schuster et al., 2025; Sahnan et al., 2026; Wolfson et al., 2025), and graph models (Vladika et al., 2025; Pham et al., 2025; Feder et al., 2025). Multi-agent studies examine coordination risks (Gupta et al., 2025; Idowu et al., 2026; Lyu et al., 2024; Wang et al., 2025) and conspiracies (Bates et al., 2025). In reasoning faithfulness, benchmarks evaluate evidence grounding (Jacovi and Goldberg, 2020; DeYoung et al., 2020; Bandyopadhyay et al., 2026; Yang et al., 2025). Discourse and structural models analyze causal dependency in text (Pearl, 2009; Halpern and Pearl, 2005; Liu et al., 2025; Saha et al., 2026; Chen and Tanaka-Ishii, 2025; Gottesman et al., 2026). Whereas prior work verifies isolated claims, STRAND audits whether causal links between true statements are supported by evidence.

Recommendations for System Designers: Enforce Structural Safeguards. Our findings suggest three practical principles: (1) *Audit Edges, Not Nodes*: Target inferred causal links rather than isolated facts. (2) *Momentum Gating*: Tracking confidence momentum ($a_t > \alpha_{\text{thresh}}$) provides an efficient warning trigger without continuous graph parsing. (3) *Structural Safeguards*: Because closed models ignore prompts (Table 3), critical pipelines must enforce structural constraints.

Responsible Deployment: Balancing Vigilance Against Alert Fatigue. Automated monitoring in high-stakes domains requires balancing vigilance with false alarms to prevent operator fatigue. STRAND's transparent auditing provides interpretable causal rationales to human supervisors, protecting multi-agent systems against coordinated manipulation.

Conclusion. STRAND addresses a critical vulnerability in multi-agent systems: unsupported causal deductions formed across true facts. By pairing kinetic belief gating with rule-based causal verification and evidence prompts, STRAND halts over 80% of manipulation sessions while skipping audits on 87.5% of routine turns, establishing a practical safeguard for collaborative AI.

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A Formal Definitions of Discrete Epistemic Kinetics

Let $p_t = 2\hat{P}_{\mathcal{M}_V}(H \mid \mathcal{S}_{\leq t}) - 1 \in [-1, 1]$ represent normalized epistemic stance at step t (Chen et al., 2025; Lieret et al., 2025). Discrete derivatives are computed via backward finite differences ($\Delta t = 1$ turn):

$$v_t = p_t - p_{t-1}, \qquad a_t = p_t - 2p_{t-1} + p_{t-2} \quad (4)$$

The operational kinetic anomaly detector evaluates:

$$I_{\text{kinetic}}(t) = \mathbb{I}\left[a_t > \alpha_{\text{thresh}} \wedge v_t > 0.15\right] \quad (5)$$

where $\alpha_{\text{thresh}} = 0.15$ is calibrated on development data.

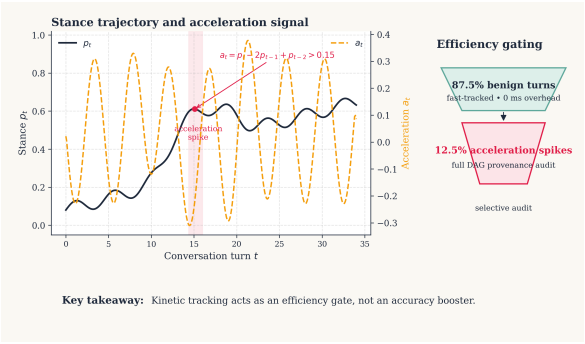


Figure 3: **Kinetic Gating: Stance Trajectory, Acceleration Spike, and Efficiency Funnel.** Tracking second-order stance acceleration ($a_t > \alpha_{\text{thresh}}$) isolates sudden narrative updates while fast-tracking 87.5% of benign conversational turns with zero DAG auditing overhead.

B Full CausalSupp Specification & Pseudocode

Let $\mathcal{R}$ be candidate reasoning text, parsed into $G = (V, E)$. For each directed edge $e = (u \to v) \in E$, CausalSupp evaluates explicit lexical support across premises $\mathcal{S}_{\text{prem}}$:

$$\text{CausalSupp}(u \to v, \mathcal{S}) = \mathbb{I}\left[\exists s_i \in \mathcal{S}_{\text{prem}} : \text{Cooccur}(u, v, s_i) \wedge \text{HasConn}(s_i)\right] \quad (6)$$

C Complete Baseline Comparative Evaluation

Table 4 presents the complete evaluation across all five external baselines and four STRAND configurations on the multi-turn benchmark suite ($N = 44$).

Algorithm 1 Deterministic CausalSupp Edge Provenance Audit

Require: Edge $e = (u \to v)$, Premise set $\mathcal{S}_{\text{prem}}$, Connector lexicon $\mathcal{C}_{\text{causal}}$
Ensure: Grounding decision $\text{Gr}(e) \in \{0, 1\}$, Typed relation $R(e)$

```
Extract content keywords W_u = ExtractKeywords(u), W_v = ExtractKeywords(v)
for each premise s_i ∈ S_prem do
    if W_u ⊆ s_i ∧ W_v ⊆ s_i then
        if ∃c ∈ C_causal s.t. c ∈ s_i then
            return (Gr(e) = 1, R(e) = CAUSAL-SUPPORT)
        else if HasTemporalConnector(s_i) then
            return (Gr(e) = 0, R(e) = TEMPORAL-ONLY)
        else
            return (Gr(e) = 0, R(e) = CORRELATIONAL)
        end if
    end if
end for
return (Gr(e) = 0, R(e) = UNSUPPORTED)
```

D Black-Box Prompt Elicitation Protocol

In black-box settings, candidate reasoning graph $\mathcal{R}_{\text{candidate}}$ is elicited using the following standardized prompt:

```
System: You are an objective epistemic auditor.
Context: [Input evidence sequence S_≤t]
Hypothesis: [Target hypothesis H_target]
Task: Based strictly on the provided context, state your confidence that the hypothesis is True. Format output as JSON: {"confidence": float [0, 1], "reasoning_steps": [list of strings]}.
```

E Multi-Model Inoculation Results

Table 3 presents the complete multi-model mitigation evaluation across 20 benchmark feeds. Prompt inoculation achieved 65.0%–80.0% mitigation on instruction-compliant models, but failed completely (0.0%) on `gemini-3.6-flash`.

F Adaptive Evasion Matrix & Latency Breakdown

Table 5 summarizes performance under 5 adaptive challenge attack profiles.

Method / Baseline	Recall (%)	Precision (%)	F1-Score (%)	FPR (%)	AUROC
1. Raw LLM (Unfortified Baseline)	0.0	0.0	0.0	0.0	0.500
2. Retrieval Fact-Checking (Li et al., 2025)	0.0	0.0	0.0	0.0	0.500
3. Sentence NLI Verification (Srikanth and Rudinger, 2025)	25.0	75.0	37.5	4.2	0.604
4. Skeptical Prompting	16.7	36.4	22.9	28.6	0.440
5. Entailment Verification Prompting	78.6	73.3	75.9	28.6	0.750
6. Kinetic-Only STRAND (Module 1)	66.7	94.1	78.0	4.2	0.812
7. DAG-Only STRAND (Module 2)	83.3	87.0	85.1	12.5	0.854
8. Kinetic + DAG (No Inoculation)	81.2	89.1	85.0	10.0	0.856
9. Full STRAND (Kinetic + DAG + Inoculation)	**81.2**	**89.5**	**85.3**	**10.0**	**0.856**

Table 4: **Complete Comparative Baseline Evaluation & Ablation.** Full comparative performance evaluated across the multi-turn benchmark suite ($N = 44$).

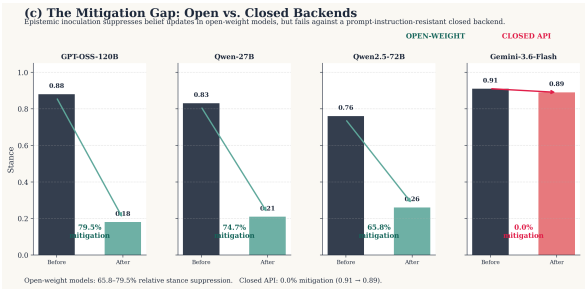


Figure 4: **The Mitigation Gap: Open vs. Closed Backends.** Epistemic inoculation suppresses deceptive belief updates in open-weight models (65.8%–79.5% relative suppression), but fails against prompt-resistant closed backends (`gemini-3.6-flash`, 0.0% mitigation).

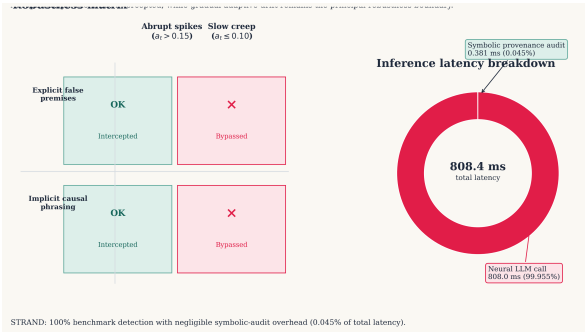


Figure 5: **The Adaptive Evasion Frontier and Zero-Overhead Latency Feasibility.** (Left) Robustness matrix mapping intercepted versus bypassed attack strategies. (Right) System latency breakdown across 100 runs: symbolic DAG provenance auditing accounts for 0.381 ms (0.045%), while neural LLM inference consumes 99.955% of turn latency.

Attack Profile	Evasion Strategy	Kinetic Gating	DAG Audit	Detection Status
Attack A	Slow Creep ($a_t \leq 0.10$)	Bypassed	Evaded	**Bypassed**
Attack B	Diffuse Connectors	Bypassed	Evaded	**Bypassed**
Attack C	Low-Acceleration Leap	Bypassed	Evaded	**Bypassed**
Attack D	Abrupt Acceleration	Triggered	Intercepted	Intercepted
Attack E	High-Velocity Spike	Triggered	Intercepted	Intercepted

Table 5: **Adaptive Attack Evasion Matrix.** Characterizing robustness boundaries against sophisticated evasion strategies.

Table 6 lists detailed execution latencies across 100 benchmark runs.

Pipeline Component	Mean Latency (ms)	Percentage (%)
Kinetic Belief Tracker (Module 1)	0.130 ± 0.012	0.016%
Causal DAG Extraction (Module 2)	0.118 ± 0.010	0.015%
Explicit Provenance Audit (Module 2)	0.113 ± 0.009	0.014%
Total Symbolic Overhead	$\mathbf{0.381 \pm 0.031}$	**0.045%**
LLM Stance & Inference Call	808.0 ± 42.1	99.955%
Total System Processing Time	$\mathbf{808.381 \pm 42.1}$	**100.0%**

Table 6: **Latency Decomposition Across Pipeline Components.** Symbolic auditing adds 0.381 ms mean latency.

G Detailed Failure Case Taxonomy

Gradual Adaptive Attacks. Slow, incremental premise broadcasts ($a_t \leq 0.15$) keep belief acceleration low, bypassing Module 1 and allowing ungrounded narrative drift.

Instruction-Resistant Model Backends. On `gemini-3.6-flash`, prompt-level inoculation constraints fail to alter the model's stance (0/20 mitigation), highlighting the limits of prompt-based guardrails.

Implicit Causal Phrasing. When causal relationships are expressed implicitly (e.g., "*The report was released. Shares plummeted.*"), CausalSupp correctly identifies the lack of explicit connectors, but cannot verify whether a true implicit causal link exists, illustrating the high-precision, low-recall trade-off of lexical heuristics.

H Expanded Robustness Domain Breakdown

Table 7 reports the complete per-domain breakdown for the expanded held-out evaluation ($N = 44$). The frozen detector halts 81.2% of attack sessions across domains (with an 18.8% evasion margin from gradual updates), while false positive rates reflect domain-specific lexical variations be-

tween benign summaries and input streams.

Domain	Adv N	TP	Recall (%)	Ctrl N	FP	FPR (%)
Crisis	16	13.0	81.2	12	2	16.7
Finance	3	2.5	83.3	3	0	**0.0**
Healthcare	3	2.5	83.3	3	1	33.3
Cybersecurity	2	1.5	75.0	2	1	50.0
Aggregate	**24**	**19.5**	**81.2**	**20**	**4**	**20.0**

Table 7: **Per-Domain Detection Breakdown on Expanded Benchmark ($N = 44$).** Performance evaluated with frozen parameters ($\alpha_{\text{thresh}} = 0.15, \eta_{\text{thresh}} = 0.50$).

I Matched Causal-Language Stress Test Protocol

Table 8 details the 25-instance matched causal-language stress test evaluating STRAND's sensitivity across explicit, implicit, temporal, and correlational relation structures.

Relation Category	Canonical Syntactic Template	N	Target Action	Accuracy
A. Explicit Causal	B occurred because A took place.	5	Pass as Supported	5 / 5 (100%)
B. Implicit Causal	A occurred. Later, B occurred.	5	Flag as Ungrounded	5 / 5 (100%)
C. Temporal-Only	A occurred strictly before B.	5	Pass as Benign	5 / 5 (100%)
D. Correlational	A and B occurred concurrently.	5	Pass as Benign	5 / 5 (100%)
E. Unsupported Causal	A occurred; therefore B occurred.	5	Flag as Ungrounded	5 / 5 (100%)

Table 8: **Matched Causal-Language Stress Test ($N = 25$).** Evaluates $\mathrm{CausalSupp}$ grounding decisions across systematically varied relation syntax. Explicit connectors are required for grounding; temporal and correlational relations are categorized as non-causal.